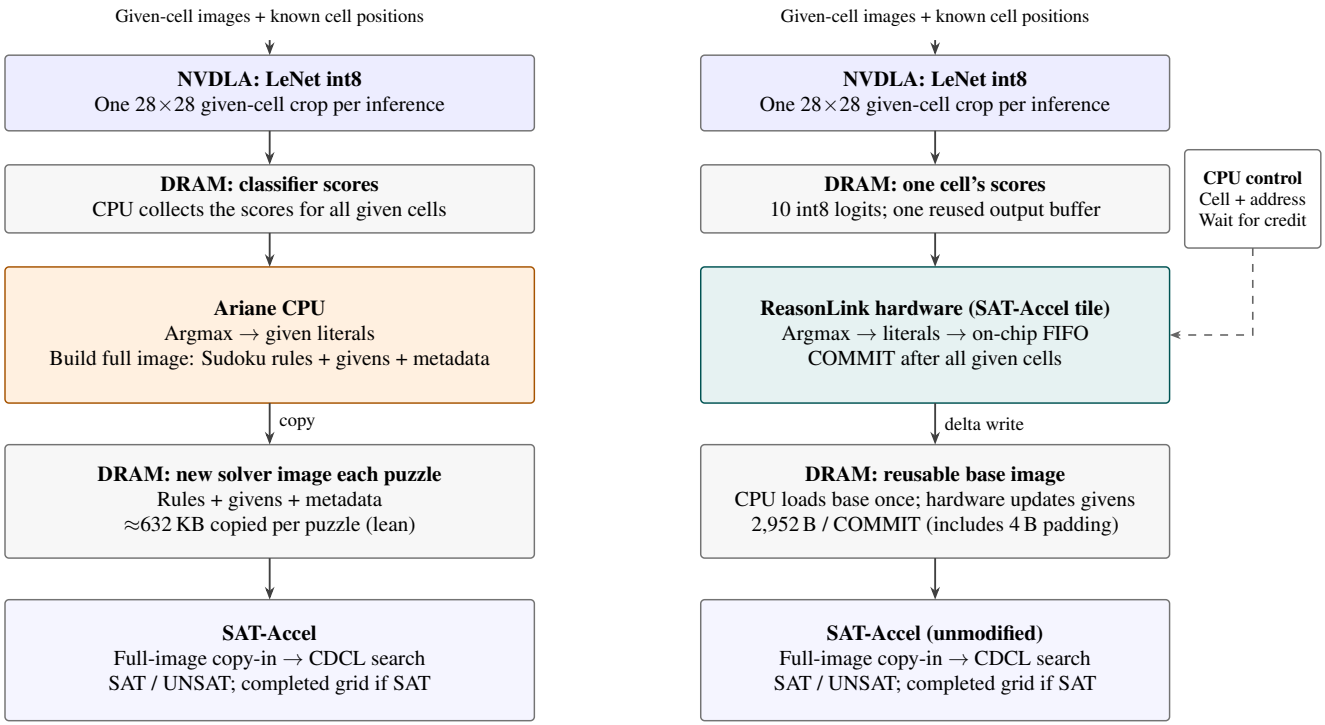
(a) Full-image baseline
Given-cell images + known cell positions
NVDLA: LeNet int8
One 28×28 given-cell crop per inference
DRAM: classifier scores
CPU collects the scores for all given cells
Ariane CPU
Argmax → given literals
Build full image: Sudoku rules + givens + metadata
copy
DRAM: new solver image each puzzle
Rules + givens + metadata
≈632 KB copied per puzzle (lean)
SAT-Accel
Full-image copy-in → CDCL search
SAT / UNSAT; completed grid if SAT
Serial: prepare i, then solve i.
Double-buffered: prepare i+1 while solving i;
two image buffers, same bytes copied per puzzle.
(b) ReasonLink Milestone 1: base in DRAM
Given-cell images + known cell positions
NVDLA: LeNet int8
One 28×28 given-cell crop per inference
DRAM: one cell's scores
10 int8 logits; one reused output buffer
CPU control
Cell + address
Wait for credit
ReasonLink hardware (SAT-Accel tile)
Argmax → literals → on-chip FIFO
COMMIT after all given cells
delta write
DRAM: reusable base image
CPU loads base once; hardware updates givens
2,952 B / COMMIT (includes 4 B padding)
SAT-Accel (unmodified)
Full-image copy-in → CDCL search
SAT / UNSAT; completed grid if SAT
Remaining cost: full-image reads on every solve.
Milestone 2: keep the base on chip and
roll back assignments and learned clauses.